ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PRACTICE EXAM 2

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 8 strands: Life and Living (a photosynthesis investigation, respiration, interactions, energy pyramids, micro-organisms and disease), Matter and Materials (changes of state, gas pressure, density, atomic structure, chemical reactions), Energy and Change (static electricity, current in series and parallel circuits, resistance, effects of current, light) and Planet Earth and Beyond (planet data, the Sun, telescopes and stars).

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL calculations with formulae and units. A non-programmable calculator may be used.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 In which part of a plant cell does photosynthesis take place?

A Nucleus B Cell wall C Vacuole D Chloroplasts

1.2 Respiration releases energy in the …

A cell wall. B mitochondria. C chloroplasts. D nucleus.

1.3 An animal that eats only plants is called a …

A carnivore. B omnivore. C herbivore. D decomposer.

1.4 Which micro-organism is used to make bread rise?

A Yeast B Virus C Algae D Cholera bacteria

1.5 The particles in a liquid …

A vibrate in fixed positions. B are close together but can slide past each other.

C are very far apart. D do not move at all.

1.6 The chemical formula for carbon dioxide is …

A CO B C_2O C C_2O_2 D CO_2

1.7 The positively charged particle in the nucleus of an atom is the …

A proton. B electron. C neutron. D molecule.

1.8 The SI unit of electric current is the …

A volt. B ohm. C ampere. D watt.

1.9 Light travels fastest through …

A water. B glass. C a vacuum. D air.

1.10 The largest planet in our solar system is …

A Jupiter. B Saturn. C Earth. D Neptune.

(10)

QUESTION 2: MATCHING COLUMNS

Choose a description from COLUMN B that matches the term in COLUMN A. Write only the letter (A–L) next to the question number, e.g. 2.11 M.

COLUMN A	COLUMN B
2.1 Glucose	A An increase in size when a substance is heated
2.2 Predator	B A huge group of billions of stars held together by gravity
2.3 Antibiotic	C The sugar made by plants during photosynthesis
2.4 Molecule	D A particle in the nucleus that has no charge
2.5 Expansion	E An animal that hunts and eats other animals
2.6 Neutron	F A material that does not let any light through
2.7 Conductor	G Two or more atoms chemically joined together
2.8 Opaque	H The path of one object around another object in space
2.9 Galaxy	I A material that allows an electric current to flow easily
2.10 Orbit	J A medicine that kills bacteria
	K A particle with a negative charge
	L A material that lets some light through

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 Grade 8 learners placed a water plant (pondweed) in a beaker of water and counted the bubbles of gas it released in one minute. They moved a lamp to different distances from the beaker.

Distance of lamp from plant (cm)	10	20	30	40
Number of bubbles per minute	40	24	12	6

3.1.1 Write down a suitable aim for this investigation. (1)

3.1.2 Identify the independent variable. (1)

3.1.3 Identify the dependent variable. (1)

3.1.4 Name the gas in the bubbles. (1)

3.1.5 Write a conclusion for this investigation. (2)

3.1.6 Name ONE variable that must be kept the same. (1)

3.2 Write the word equation for respiration. (3)

3.3 (a) What is a habitat? (b) What is meant by competition between organisms? (c) Give an example of two organisms that compete, and say what they compete for. (3)

3.4 Explain why there are always fewer carnivores than herbivores in an ecosystem. (2)

3.5 (a) Name the FOUR main groups of micro-organisms. (b) Give TWO ways to prevent the spread of diseases caused by micro-organisms. (c) Name ONE disease caused by bacteria. (5)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 (a) Describe what happens to the particles of ice when it melts. (b) Name the change of state from a gas to a liquid. (c) Name the change of state from a solid directly to a gas. (4)

4.2 Explain, using the particle model, why a blown-up balloon may burst if it is left in a hot car. (3)

4.3 Honey (density 1,4 g/cm³), water (1,0 g/cm³) and cooking oil (0,9 g/cm³) are poured into a tall glass. (a) In what order will the liquids settle, from the bottom to the top? (b) Where will a cork (0,24 g/cm³) float? (3)

4.4 A helium atom has 2 protons, 2 neutrons and 2 electrons. (a) Where in the atom are the protons and neutrons found? (b) What is the charge of an electron? (c) Why does the atom have no overall charge? (3)

4.5 Write down the chemical symbols for carbon and chlorine. (2)

4.6 Magnesium ribbon burns in oxygen with a bright white flame to form a white powder, magnesium oxide. (a) Write a word equation for this reaction. (b) Is this a chemical change or a physical change? Give a reason. (4)

4.7 Give ONE difference between a mixture and a compound. (1)

[20]

QUESTION 5: ENERGY AND CHANGE

5.1 (a) Explain why you sometimes feel a small shock when you touch a metal door handle after walking on a carpet. (b) Name the device that protects tall buildings from lightning damage. (3)

5.2 (a) In a series circuit, an ammeter next to the cell reads 0,4 A. What will an ammeter between the two bulbs read? (b) In a parallel circuit with two branches, the total current is 0,6 A and the current in one branch is 0,4 A. Calculate the current in the other branch. (c) Explain your answer to (b). (4)

5.3 (a) Name TWO factors that affect the resistance of a wire. (b) Why are the heating elements of stoves made of nichrome wire? (3)

5.4 Name the effect of an electric current used in: (a) an electric bell; (b) electroplating a spoon with silver; (c) a toaster. (3)

5.5 Light

5.5.1 Explain the difference between transparent, translucent and opaque materials. Give ONE example of each. (3)

5.5.2 Explain how a rainbow forms. (2)

5.5.3 What colour will a green leaf appear in pure red light? Explain. (2)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 Study the table about some planets.

Planet	Average distance from the Sun (million km)	Time to orbit the Sun
Mercury	58	88 days
Earth	150	365 days
Mars	228	687 days
Jupiter	778	12 years
Neptune	4 495	165 years

6.1.1 Describe the relationship between a planet's distance from the Sun and the time it takes to orbit the Sun. (2)

6.1.2 Which planet's year is closest to two Earth years? (1)

6.1.3 Sunlight takes about 8 minutes to reach the Earth. Estimate how long it takes sunlight to reach Mars. Show your working. (2)

6.1.4 Why is Neptune extremely cold? (1)

6.2 Give TWO facts about the Sun. (2)

6.3 Which TWO planets have no moons? (2)

6.4 (a) What is the difference between an optical telescope and a radio telescope? (b) Why are some telescopes placed in space instead of on Earth? (4)

6.5 What is a constellation? Name ONE constellation that can be seen from South Africa. (2)

6.6 What is the name of the huge cloud of gas and dust in which stars are formed? (1)

6.7 Why do we always see the same side of the Moon from Earth? (2)

6.8 Name ONE galaxy, other than the Milky Way, that can be seen without a telescope from South Africa. (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER